Machine Learning Basics to Advanced - Skill IQ Practice Question Bank

Medium, Hard, and Very Hard MCQs with answer key and explanations

Purpose: Build Skill IQ-style readiness using original scenario-based MCQs. These are not official Pluralsight exam questions or dumps.

Difficulty mix: Medium questions test core knowledge; Hard questions test applied decision-making; Very Hard questions test architecture, edge cases, trade-offs, and production judgement.

Focus areas

- Supervised, unsupervised, and semi-supervised learning
- Bias-variance, regularization, train/validation/test design, and leakage
- Classification, regression, clustering, metrics, calibration, and model selection
- Feature engineering, dimensionality reduction, trees, ensembles, SVMs, and neural networks
- Deployment, monitoring, drift, fairness, and MLOps concepts

How to use this bank

Do the questions timed first, then review the answer key. For Skill IQ-style prep, focus on why the correct answer is better than plausible alternatives.

Source scope

Aligned with public Pluralsight path descriptions and general industry knowledge. Topics emphasize breadth, applied reasoning, and production trade-offs rather than memorized question dumps.

Questions

Q01 - Medium - Problem framing

A bank wants to predict whether a loan applicant will default within 12 months. Which ML framing is most appropriate?

- **A.** Unsupervised clustering
- **B.** Binary classification
- **C.** Regression with continuous target
- **D.** Reinforcement learning

Q02 - Medium - Data split

Why should a test set be kept separate until final evaluation?

- **A.** To reduce training time
- **B.** To tune hyperparameters more aggressively
- **C.** To estimate generalization on unseen data
- **D.** To increase model capacity

Q03 - Medium - Metrics

For a heavily imbalanced fraud dataset where catching fraud matters more than avoiding false alarms, which metric is often more informative than accuracy?

- **A.** Mean squared error
- **B.** Recall or precision-recall curve
- **C.** R-squared
- **D.** Silhouette score

Q04 - Medium - Bias-variance

A decision tree fits training data almost perfectly but performs poorly on validation data. What is the most likely issue?

- **A.** High bias
- **B.** High variance
- **C.** Underfitting
- **D.** Insufficient feature scaling

Q05 - Medium - Regularization

What is the main purpose of L2 regularization in linear models?

- **A.** Force all coefficients to exactly zero
- **B.** Penalize large coefficients to reduce overfitting
- **C.** Convert regression into clustering
- **D.** Increase training error intentionally without affecting generalization

Q06 - Medium - Preprocessing

Which algorithm is most sensitive to feature scaling?

- **A.** Decision tree
- **B.** Random forest
- **C.** k-nearest neighbors
- **D.** Rule-based lookup

Q07 - Medium - Regression

What does mean absolute error measure?

- **A.** Average absolute prediction error
- **B.** Average squared prediction error
- **C.** Fraction of correctly classified points
- **D.** Cluster compactness

Q08 - Medium - Classification

In logistic regression, the model output before thresholding is best interpreted as what?

- **A.** A cluster label
- **B.** A probability or score for the positive class
- **C.** A principal component
- **D.** A reward signal

Q09 - Medium - Unsupervised learning

Which task is unsupervised?

- **A.** Predicting house price from features
- **B.** Classifying emails as spam or not spam
- **C.** Grouping customers by behavior without labels
- **D.** Predicting churn from historical churn labels

Q10 - Medium - PCA

What is the goal of PCA?

- **A.** Find directions of maximum variance and reduce dimensionality
- **B.** Create labels for supervised learning
- **C.** Increase feature count to avoid underfitting
- **D.** Guarantee causal explanations

Q11 - Medium - Trees

Why are decision trees often considered interpretable compared with neural networks?

- **A.** They always have higher accuracy
- **B.** They use explicit feature splits that can be traced
- **C.** They require GPUs
- **D.** They never overfit

Q12 - Medium - Validation

What is k-fold cross-validation mainly used for?

- **A.** Encrypting the dataset
- **B.** Estimating model performance more robustly across splits
- **C.** Replacing all test sets
- **D.** Converting labels into embeddings

Q13 - Hard - Leakage

A model predicting hospital readmission includes a feature named discharge_followup_completed, which is only known after discharge. Why is this risky?

- **A.** It is categorical
- **B.** It may leak future information unavailable at prediction time
- **C.** It reduces feature count
- **D.** It must be scaled with min-max scaling

Q14 - Hard - Metrics

Two classifiers have the same ROC-AUC on a rare disease dataset, but one has much higher precision at the operating recall. Which should be preferred if false positives are costly?

- **A.** The one with higher precision at the chosen recall
- **B.** The one with lower training loss only
- **C.** The one with more parameters
- **D.** The one trained on fewer examples

Q15 - Hard - Calibration

A classifier is accurate but consistently assigns 0.95 probability to events that occur only 70 percent of the time. What is the issue?

- **A.** Poor calibration
- **B.** Perfect recall
- **C.** High silhouette score
- **D.** Good regularization

Q16 - Hard - Model selection

A linear model underfits even after reducing regularization. Which next step is most reasonable?

- **A.** Add non-linear features or try a more expressive model
- **B.** Delete the validation set
- **C.** Increase test leakage
- **D.** Use accuracy for a regression task

Q17 - Hard - Imbalance

Why can oversampling the minority class before train-test splitting cause leakage?

- **A.** It makes the dataset smaller
- **B.** Duplicate or synthetic variants can appear in both train and test sets
- **C.** It prevents feature scaling
- **D.** It removes all outliers

Q18 - Hard - SVM

What does the kernel trick allow an SVM to do?

- **A.** Compute distances in the original input only
- **B.** Use implicit high-dimensional feature spaces without explicitly constructing them
- **C.** Avoid all regularization
- **D.** Train without labels

Q19 - Hard - Ensembles

Why does bagging, as used in random forests, often reduce variance?

- **A.** It trains identical models on identical data
- **B.** It averages diverse models trained on bootstrap samples
- **C.** It converts regression to classification
- **D.** It removes all weak features

Q20 - Hard - Boosting

Gradient boosting primarily builds new learners to do what?

- **A.** Correct errors or residuals of previous learners
- **B.** Replace the loss function with accuracy
- **C.** Randomly ignore all hard examples
- **D.** Guarantee zero training loss

Q21 - Hard - Neural networks

What problem do residual connections mainly help with in deep networks?

- **A.** They make labels unnecessary
- **B.** They improve gradient flow and make deeper networks easier to train
- **C.** They remove all nonlinearities
- **D.** They force probabilities to be calibrated

Q22 - Hard - Optimization

If training loss is noisy and unstable with mini-batch gradient descent, which change often helps?

- **A.** Increase learning rate dramatically
- **B.** Lower learning rate or use an adaptive optimizer
- **C.** Remove validation data
- **D.** Train only on the test set

Q23 - Hard - Feature engineering

Why might target encoding need careful cross-validation?

- **A.** It requires GPUs
- **B.** It can leak label information if category statistics are computed using the same row's target
- **C.** It only works for images
- **D.** It prevents overfitting automatically

Q24 - Hard - Drift

A deployed model's input feature distribution changes while the relationship between features and target remains stable. What type of drift is this most likely?

- **A.** Covariate drift
- **B.** Concept drift
- **C.** Label leakage
- **D.** Perfect calibration

Q25 - Hard - Interpretability

What is one limitation of SHAP or feature attribution methods?

- **A.** They always prove causality
- **B.** They may explain model behavior, not the true causal process
- **C.** They cannot be used with tree models
- **D.** They require labels at inference

Q26 - Very Hard - Evaluation design

A time-series demand model is validated using random k-fold CV and performs extremely well, but fails in production. What is the most likely validation flaw?

- **A.** Random CV can leak future temporal patterns into training folds
- **B.** The model had too few features because time-series models need all possible features
- **C.** The test set was too large
- **D.** Regression cannot be used for demand forecasting

Q27 - Very Hard - Thresholds

A fraud model outputs calibrated probabilities. The business wants to review only 1,000 cases per day. What should determine the classification threshold?

- **A.** Default threshold of 0.5 always
- **B.** The score cutoff that selects the top 1,000 expected-risk cases under capacity constraints
- **C.** The threshold that maximizes training accuracy
- **D.** A random threshold to avoid bias

Q28 - Very Hard - Causality

A churn model finds that customers who receive retention calls are more likely to churn. Why should this not be interpreted as calls causing churn?

- **A.** ML models cannot use phone-call features
- **B.** The feature may reflect intervention targeted at already high-risk customers
- **C.** Classification models cannot detect correlations
- **D.** The dataset must be unsupervised

Q29 - Very Hard - Fairness

A model excludes a protected attribute but uses postcode, income, and school as features. Why can fairness issues remain?

- **A.** Protected attributes are never relevant
- **B.** Proxy variables may encode protected-class information
- **C.** Only neural networks have fairness issues
- **D.** Fairness is solved by removing all categorical columns

Q30 - Very Hard - Nested CV

Why use nested cross-validation for model comparison with hyperparameter tuning?

- **A.** Outer folds estimate generalization after inner-fold tuning, reducing selection bias
- **B.** It makes the model unsupervised
- **C.** It eliminates the need for train data
- **D.** It guarantees the best production model

Q31 - Very Hard - Learning curves

A learning curve shows high training error and high validation error, both plateauing. What is the best diagnosis?

- **A.** High variance only
- **B.** High bias or insufficient model capacity/features
- **C.** Data leakage
- **D.** Threshold miscalibration only

Q32 - Very Hard - MLOps

A model's offline AUC remains stable, but business KPIs degrade after deployment. What should be investigated first?

- **A.** Only retrain with more epochs
- **B.** Data pipeline parity, decision thresholds, delayed labels, and operational feedback loops
- **C.** Remove monitoring to reduce latency
- **D.** Switch from Python to Java

Q33 - Very Hard - Clustering

Why can a high silhouette score still be misleading?

- **A.** It always means labels are correct
- **B.** It measures geometric separation, not whether clusters are useful or semantically meaningful
- **C.** It cannot work with Euclidean distance
- **D.** It requires a neural network

Q34 - Very Hard - Regularization and bias

Increasing L1 regularization in a sparse linear model is most likely to do what?

- **A.** Increase coefficient sparsity and potentially increase bias
- **B.** Increase every coefficient equally
- **C.** Remove the need for validation
- **D.** Make a non-linear kernel automatically

Q35 - Very Hard - Production monitoring

Which signal best detects concept drift when labels are delayed but eventually available?

- **A.** Only CPU utilization
- **B.** Change in model performance once delayed labels arrive, combined with input and prediction drift monitors
- **C.** Number of rows in the training dataset only
- **D.** Code formatting changes

Q36 - Very Hard - Deep learning

Why can batch normalization behave differently during training and inference?

- **A.** It uses batch statistics during training and running estimates during inference
- **B.** It disables all weights during inference
- **C.** It converts supervised learning into clustering
- **D.** It requires one-hot labels at inference

Answer Key with Explanations

Q01. B - Binary classification
Problem framing / Medium
The target is a yes/no outcome, so this is binary classification.

Q02. C - To estimate generalization on unseen data
Data split / Medium
The test set should simulate unseen future data. Using it during tuning leaks evaluation information.

Q03. B - Recall or precision-recall curve
Metrics / Medium
Accuracy can be misleading on imbalanced data. Recall and precision-recall trade-offs are more relevant.

Q04. B - High variance
Bias-variance / Medium
A very flexible tree can memorize the training data, causing high variance and poor generalization.

Q05. B - Penalize large coefficients to reduce overfitting
Regularization / Medium
L2 regularization shrinks large weights, usually reducing variance and overfitting.

Q06. C - k-nearest neighbors
Preprocessing / Medium
KNN depends on distance calculations, so features with larger scales can dominate unless scaled.

Q07. A - Average absolute prediction error
Regression / Medium
MAE is the average of absolute differences between predictions and actual values.

Q08. B - A probability or score for the positive class
Classification / Medium
Logistic regression maps a linear score through a sigmoid to produce a probability-like output.

Q09. C - Grouping customers by behavior without labels
Unsupervised learning / Medium
Clustering customers without pre-existing labels is an unsupervised task.

Q10. A - Find directions of maximum variance and reduce dimensionality
PCA / Medium
PCA projects data onto orthogonal components that capture maximum variance.

Q11. B - They use explicit feature splits that can be traced
Trees / Medium
A tree prediction can be traced through a sequence of human-readable feature decisions.

Q12. B - Estimating model performance more robustly across splits
Validation / Medium
K-fold CV trains and validates across multiple partitions, reducing dependence on one validation split.

Q13. B - It may leak future information unavailable at prediction time
Leakage / Hard
Features not available at prediction time create target leakage and unrealistic validation performance.

Q14. A - The one with higher precision at the chosen recall
Metrics / Hard
Operational threshold metrics matter. If false positives are costly, precision at the chosen recall is important.

Q15. A - Poor calibration
Calibration / Hard
Calibration measures whether predicted probabilities match observed frequencies.

Q16. A - Add non-linear features or try a more expressive model
Model selection / Hard
Persistent underfitting suggests high bias, so more expressive features or models can help.

Q17. B - Duplicate or synthetic variants can appear in both train and test sets
Imbalance / Hard
Oversampling before splitting can put near-duplicates in both sets, inflating test performance.

Q18. B - Use implicit high-dimensional feature spaces without explicitly constructing them
SVM / Hard
Kernels compute inner products in implicit feature spaces, enabling non-linear boundaries.

Q19. B - It averages diverse models trained on bootstrap samples
Ensembles / Hard
Averaging imperfectly correlated models reduces variance.

Q20. A - Correct errors or residuals of previous learners
Boosting / Hard
Boosting adds learners sequentially to reduce residual errors under a chosen loss.

Q21. B - They improve gradient flow and make deeper networks easier to train
Neural networks / Hard
Residual paths allow gradients and information to flow through deep networks more easily.

Q22. B - Lower learning rate or use an adaptive optimizer
Optimization / Hard
A high learning rate can cause unstable optimization; lower rates or adaptive optimizers can stabilize training.

Q23. B - It can leak label information if category statistics are computed using the same row's target
Feature engineering / Hard
Target encoding uses label statistics. Done incorrectly, it leaks target information into features.

Q24. A - Covariate drift
Drift / Hard
Covariate drift is a change in input distribution. Concept drift changes the target relationship.

Q25. B - They may explain model behavior, not the true causal process
Interpretability / Hard
Attributions describe model reliance, not necessarily causal relationships in the world.

Q26. A - Random CV can leak future temporal patterns into training folds
Evaluation design / Very Hard
For temporal data, validation should respect time order. Random splits can train on future data and validate on the past.

Q27. B - The score cutoff that selects the top 1,000 expected-risk cases under capacity constraints
Thresholds / Very Hard
Operational capacity should drive the threshold when only a fixed number of cases can be actioned.

Q28. B - The feature may reflect intervention targeted at already high-risk customers
Causality / Very Hard
Retention calls may be assigned to customers already likely to churn, creating confounding.

Q29. B - Proxy variables may encode protected-class information
Fairness / Very Hard
Features correlated with protected classes can act as proxies even if the protected attribute is removed.

Q30. A - Outer folds estimate generalization after inner-fold tuning, reducing selection bias
Nested CV / Very Hard
Nested CV separates tuning from performance estimation, reducing optimistic bias.

Q31. B - High bias or insufficient model capacity/features
Learning curves / Very Hard
Both errors high and close together usually indicate underfitting/high bias.

Q32. B - Data pipeline parity, decision thresholds, delayed labels, and operational feedback loops
MLOps / Very Hard
Offline metrics may not reflect production behavior due to pipeline, threshold, label, or workflow changes.

Q33. B - It measures geometric separation, not whether clusters are useful or semantically meaningful
Clustering / Very Hard
Cluster metrics may not align with business meaning or downstream utility.

Q34. A - Increase coefficient sparsity and potentially increase bias
Regularization and bias / Very Hard
L1 can drive coefficients to zero, improving sparsity but possibly increasing bias if overdone.

Q35. B - Change in model performance once delayed labels arrive, combined with input and prediction drift monitors
Production monitoring / Very Hard

Concept drift involves changed feature-target relationships, requiring eventual labels to confirm performance shifts.

Q36. A - It uses batch statistics during training and running estimates during inference
Deep learning / Very Hard
Batch norm normalizes with batch statistics while training and stored running statistics at inference.

Last-minute revision checklist

- Explain supervised, unsupervised, regression, classification, and clustering clearly.
- Know train/validation/test splitting, leakage, cross-validation, and temporal validation.
- Choose metrics for imbalanced classification, regression, ranking, and clustering.
- Diagnose high bias vs high variance using learning curves.
- Understand regularization, feature scaling, PCA, trees, ensembles, SVMs, and neural network basics.
- Know deployment risks: drift, monitoring, calibration, fairness, and feedback loops.

Suggested timed drill

Pick 25 mixed questions and give yourself 15 minutes. Then review every wrong or guessed answer and write one sentence explaining the underlying concept.